YEAR 4 MATHEMATICS & PHYSICS MASTER'S STUDENT, WARWICK UNIVERSITY (2023-2027)

ALFIE KUNZ

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KEY SKILLS: **Programming**: C#, Python, VB.NET, HLSL. **Tools & Frameworks**: Unity Engine, Git, WinForms, LaTeX, Tikz, Linux, SSH.

Core Competencies: High Performance Computing, Computational Fluid Dynamics, (GPU) Agent-based Modelling, Mathematical Modelling, Technical Writing & Communication, Machine Learning (Neural Networks).

EDUCATION

UNIVERSITY OF WARWICK: Full-Time Student of Mathematics & Physics (MMathPhys) **2023 – 2027**

- On track for a high First-Class Honours degree (**>80%**), with notable achievements being my Year 1 Grade of **88.0%** ('Top 5' in the Cohort).
- Years 1-2 high module breadth: Mathematical Analysis, Applied Modelling and Algebra, Mathematics of Asymptotics, Physics of Quantum Mechanics, Applied Mathematics, Fluids and Electrodynamics, Computational Modelling, Programming, and Algorithms.
- 3rd Year Specialism through modules such as Computational Modelling, Maths Essay, Mathematical Modelling of PDEs, and Fluid Dynamics.
- Ongoing 4th year research into Fluid Dynamics, Computational Modelling using PDEs, and Technical Writing (Liquid Drop collisions).

BECKFOOT SCHOOL: A-Level & GCSE Student **2016 – 2023**

- A-Level Grades: **A*A*A*A*A** for Mathematics, Further Mathematics, Physics, Computer Science, EPQ (Astrophysics-based: "How Does a Neutron Star Remain a Stable Entity?"). #1 in Cohort. GCSE Grades: Ten Grade 9s, as well as a Distinction in Music BTEC.
- Highlights & Achievements: Founder & Treasurer of the School Chess Club, served as **featured Alumni** for Open Days & Showcases.

PROFILE: Proactive computational researcher, ranked in the **'Top 5' of my university cohort**, specialising in fluid dynamics, numerical simulations, and emergent behaviours in biophysics. Proven success in translating cutting-edge mathematical theory into **commercial-grade, high-performance software**. Inventive science communicator, puzzle designer, and believer in the idea that the most complex behaviours stem from elegantly simple rules.

KEY STRENGTHS: **Charismatic Leadership** & Operational Initiative**, Creative Reasoning** & Problem Solving, **Technical & Interpersonal Communication**, Mentorship & Coaching, Adaptability & High-Pressure Composure, **Self-Direction**.

PROJECT EXPERIENCE

Drop Collisions, and their Collective Behaviour (in Collaboration with the **Met Office**) **2026 – 2027**

University of Warwick (Associated with Prof James Sprittles, under the PX402 Physics Project Module)

- Currently undertaking a research project on computationally predicting the boundaries between bouncing and merging of liquid drops, by developing **a reduced-order 'boundary integral' numerical model**. These multiphase flow dynamics become crucial for studying cloud formation, and thus weather and climate modelling. Project in collaboration with partners including the **Met Office**, **NCAR**, and **Syngenta**.

APIC Computational Fluid Dynamics (CFD) Research Project & Simulation **2025 – 2026**

University of Warwick (Associated with Prof Radu Cimpeanu, under the MA369 Essay Module) (C#, HLSL, Python, Unity)

- Identified a **novel Reynolds Number value of 5420** for the APIC CFD regime, in a Taylor-Green Vortex – justifies the report as a worthwhile contribution to the industry. Achieved by self-directing original CFD research, applying cutting-edge technologies in novel scenarios to investigate high-performance, momentum-conserving vortices.
- Independent derivation and implementation of a real-time **Eulerian-Lagrangian APIC system** (C#, Unity), using techniques such as Conjugate Gradient Method with MICT-0 Preconditioning, and GPU parallelism. Optimised rendering via Instanced Procedural Drawing.
- Automated post-processing, data analysis, and visualisation pipelines in Python.
- **Authored a rigorous, well-referenced technical essay** on my contributions to the field, using LaTeX and Tikz figures. Demonstrated strong scientific communication by **presenting my findings** to two specialist Warwick professors (with no prior exposure to the implementation).
- Currently in the process of **peer-reviewing** and grounding said essay, by submitting to the 'SIAM Undergraduate Research Online' journal.
- Achieved a mark of **87%** (the highest across all my third-year mathematics modules), with the essay itself commended for demonstrating scientific rigour and computational mathematics "**far beyond the level of a typical 3rd-year undergraduate student**".
- ***Outcome:*** *identifying, documenting, and presenting highly technical work (to two specialists) gave confidence in both* ***communicating complex ideas under scrutiny*** *(a skill I've since relied on as a Student Ambassador) and* ***academic writing*** *(providing a fantastic foundation for my upcoming dissertation).*

Chess Artificial Intelligence & Interface **2021 – Present**

Beckfoot School A-Level Project (+ Self-Motivated Continuation) (VB.NET, WinForms)

- Architected a **'Grandmaster' Elo** (~2850, as of v11.0), commercial-quality Chess AI project fully from scratch, designed around a stylish chess-playing interface (to help with, for example, training and ranking via a hand-selected puzzle database & infrastructure).
- **Awarded 100%** on the initial A-Level CS NEA; recognised by the project supervisor as the 'best he had seen' in his years of teaching.
- Robustly engineered and optimised a custom **NegaMax heuristic AI algorithm** (built around a fully bitboard architecture), using cutting-edge (and often novel) optimisation techniques such as 'Fancy Magic' bitboards, Zobrist Hashing, Internal Iterative Reductions, etc.
- Evaluated against a vast breadth of variations, by integrating models into a sophisticated, custom-made swiss-style benchmarking system with a 'plug & play' ecosystem. Includes a custom Zobrist hashing engine for robust game-ending detection, and colourful live visuals.
- Contributed an open-source, easy-to-use, and highly-customisable opening book builder (made from millions of open-source games) as a GitHub repository to support the engine, in an area lacking of high-quality, easy-to-access code. Contains features such as automatic PGN parsing, built-in move validation, and automatic negligible move pruning.
- ***Outcome:*** *demonstrated clear evidence of* ***motivated, self-directed operational initiative*** *towards a highly technical, production-grade product which I am still improving. Re-shaped my academic specialties as a performance-oriented, open-source driven programmer with* ***strong knowledge of code profiling****.*

Machine Learning & Biophysical Modelling Experiments **2023 , 2024**

Independent Research Projects (C#, HLSL, VB.NET)

- **Emergence from Evolving Slime Trails**: Simulated millions of atomic agents in real-time on the GPU (massively parallelised as a compute shader) to conduct biophysical research on agent-based physarum transport networks under a **'Patlak-Keller-Segel'**-type couple system, and their evolution into emergent ecosystems. Self-taught C# and HLSL, under the Unity Game Engine.
- **Hand-drawn Doodle & Digit Recognising Neural Network (NN)**: Achieves **~98% accuracy on the MNIST dataset, and ~92% on the Google 'Quick Draw' database** (12 categories) in under 2 minutes of training. Employs high-level ML concepts such as back propagation, regularisation, and overfitting regulation. Augments database images for strong resilience against user-drawn images.
- ***Outcome:*** *proved* ***academic prowess*** *and* ***self-directedness*** *by self-teaching languages, modules, and graphics pipelines from scratch. Provided necessary experience for third-year university modules (Computational Modelling and Neural Computing) and the technicalities of my CFD research project.*

Photography and Academic Web-based Portfolio **2025 - Present**

Independent Research Project (HTML, CSS, JS, Python)

- Overhauled and re-engineered a baseline template into a **bespoke academic and photography portfolio**, which serves as a home for both photography advertisement and career building.
- Developed custom JavaScript modules for dynamic JSON gallery rendering and hosting, AES-256 encrypted client-side delivery, custom non-linear phase modulation background, and integration with GitHub repositories.
- ***Outcome:*** *proven success in receiving contracted photography work, and CV building through easy access to my entire project library.*

OTHER AWARDS: 4-time Distinction & Gold Award Achiever in 'Bebras' Computing Challenge (**top 10% nationally**); 2-time Distinction holder in 'TCS Oxford Computing Challenge'. Two-time Winner of 'Beckfoot Young Musician of the Year' (piano).

RELEVANT EXPERIENCE & VOLUNTARY WORK

Graduate Teaching Assistant: University of Warwick **2026 – Present**

- **Selected by faculty** to design and facilitate weekly personal seminars for first-year undergraduates, **establishing technical rigour and fluency** across a variety of core modules, whilst developing **mathematical confidence** through interactive group problem solving.
- Directed formal marking of assignments with detailed, actionable feedback (representative of the differences in individual abilities), focussing not just on mathematical correctness but clarity of communication and elegance of solutions.
- Authored **open-source revision notes** and geometric illustrations to demystify abstract concepts and provide 'lightbulb' moments.
- Monitored student wellbeing and academic trajectory throughout the year, serving as a first point of contact for **pastoral support**.

Physics Student Ambassador: University of Warwick **2024 - 2026**

- Led and assisted **Live Chat events, Q&A panels, and Department & Campus Tours**, for prospective students. Rapport-building and adaptation to different perspectives (developed through training workshops) to provide an exciting, interactive insight into university life.
- ***Outcome:*** *consistently positive feedback led to me being re-hired across the rest of my degree: sharpened my ability to read a wide range of cohort's energy.*

Teaching Assistant & GCSE Tutor: Beckfoot School **2021 – 2022 , 2025**

- Mentored lower ability Year 11 students in mathematics & science, involving **one-on-one interpersonal support** with strong client communication (with development of personalised revision plans) to tackle 'maths anxiety' and providing inspirational 'lightbulb' moments.
- Shadowed staff, and worked with diverse students, to discern the strengths of approachability and patience in tackling knowledge gaps.
- **Designed and independently delivered an A-level maths lesson** on series convergence (extending the spec to introduce Ramanujan summation and Zeno paradoxes) – achieving strong student engagement, and reinforcing my ideals of inspiration via creative teaching.
- Reflected heavily on my own performance by proactively seeking and implementing feedback from experienced staff.
- ***Outcome:*** *being invited back to the department in 2025 (years after my initial tutoring role) gave me the confidence to thrive as an* ***interpersonal and science communicator****, a skill which proved vital for presenting my 3rd Year Maths Essay to two experts in the field.*

Retail Sales Associate: Whittakers Schoolwear, Shipley **2023 - 2024**

- Rapidly adapted to a **high-intensity retail environment during peak summer season**, whilst maintaining professionalism and composure. Expanded responsibilities to encompass till operations, database operations, online order management, and onboarding of employees.
- ***Outcome:*** *being trusted with duties well beyond my original scope proved my* ***adaptability*** *and* ***credibility*** *– strengths I value immensely in academic work.*

Chess Club Founder, Leader, Treasurer: Beckfoot School **2021 - 2023**

- Founded and effectively scaled a school chess club to a **weekly crowd of 70+ young people**. Showed strong initiative and handling school resources. Involved the hiring of volunteers, organising class teaching and individual mentorship.
- Established a tournament structure to enable students to represent their school in regional competitions, **proving a concrete legacy**.
- ***Outcome:*** *sculpted my* ***charismatic leadership****, effective budgeting, and asset handing. Provided the core of my energetic yet patient approach to* ***mentoring****.*

WIDER INTERESTS: Beyond my degree, I find fulfilment in creative streams such as **Music** (self-taught pianist of 16 years, performer at weddings) and **Photography** (specialist in street & landscape photography, regularly undertaking paid event work for weddings, society balls, and corporate clients). I'm at my best when connecting with new people and hosting engaging problem-solving experiences - whether it's designing & running social deduction and puzzle games for groups at my University, watching in amusement as my carefully planned mechanisms unfold, or competing in board games & escape rooms with friends. I pride myself on my **'jack of all trades'** approach to life, constantly refining my non-academic interests through recent adventures such as salsa partnerwork, learning the alto saxophone, and even Ultimate Frisbee!
